

AI-Powered Candidate Intelligence and Job Readiness Platform

Phase 1 - Candidate Profile, Gap Filling, Role Recommendation and Adaptive Assessment

Project Type: AI Recruitment / Candidate Intelligence Platform

Primary Users: Job candidates

Phase 1 Goal: Convert an unstructured resume into a complete candidate profile, understand the candidate's capabilities, recommend suitable job roles, and establish a measurable readiness benchmark.

1. Project Overview

The platform is a candidate-focused recruitment system that builds a detailed understanding of each user before recommending jobs. Instead of relying only on the information directly available in a resume, the system parses the resume, stores useful knowledge for retrieval, detects missing information, asks targeted follow-up questions, and gradually builds a structured candidate profile.

Once the profile reaches an acceptable level of completeness, the system recommends job roles based on the candidate's education, experience, projects, skills and interests. The candidate then selects preferred roles and completes an adaptive assessment. The final output is a candidate benchmark showing current readiness, strengths, weaknesses and areas that should be improved.

2. Phase 1 Objective

Phase 1 should create the complete candidate intelligence pipeline required before actual job matching and recruitment automation.

- Accept and process candidate resumes.
- Extract structured information from unstructured resume content.
- Ingest relevant resume information into a Retrieval-Augmented Generation (RAG) knowledge layer.
- Create a standard candidate information schema defining what information should ideally be available.
- Detect missing or insufficient candidate information.
- Ask contextual follow-up questions to fill important gaps.
- Build and continuously update a structured candidate profile.
- Recommend suitable job roles based on candidate data.
- Allow the candidate to select approximately 2-3 preferred roles from the recommendations.
- Run an adaptive knowledge and capability assessment for the selected roles.
- Generate measurable candidate readiness metrics and a benchmark.
- Identify what the candidate should improve to become eligible for stronger opportunities.

3. End-to-End Candidate Flow

Stage	Process	Output
1	Candidate Login / Registration	Candidate account
2	Resume Upload	Original resume
3	Resume Parsing	Structured resume data
4	RAG Ingestion	Searchable candidate knowledge
5	Completeness Analysis	Missing / weak fields
6	Gap-Filling Conversation	Additional candidate information
7	Profile Generation	Unified candidate profile
8	Role Recommendation	Suggested job roles

Stage	Process	Output
9	Role Selection	2-3 preferred roles
10	Adaptive Assessment	Knowledge and capability evidence
11	Benchmarking	Readiness metrics and improvement gaps

4. Module 1 - Resume Upload and Parsing

The candidate begins by uploading a resume, preferably in PDF or another supported document format. The resume parser extracts important information and converts it into a normalized structure that can be used by the rest of the platform.

Expected information

- Basic candidate information.
- Education and qualifications.
- Technical and non-technical skills.
- Employment and internship experience.
- Projects and project responsibilities.
- Certifications and training.
- Tools, frameworks and technologies.
- Achievements and relevant activities.
- Role titles, durations and experience levels where available.

5. Module 2 - Candidate RAG Knowledge Layer

After parsing, useful resume information should be converted into retrievable candidate knowledge. The system can chunk candidate information, generate embeddings and store the resulting representations in a vector database. Structured profile data should still be maintained separately as the authoritative candidate record.

The RAG layer is intended to help the AI retrieve evidence from the candidate's resume and later responses when generating questions, recommendations, explanations or assessments. It should not replace the structured profile database.

6. Module 3 - Candidate Information Criteria and Completeness

A standard candidate schema should define the minimum information expected before the system considers a profile sufficiently complete. Each category can have required fields, optional fields and a completeness weight.

Category	Examples
Identity / Basics	Name, contact details, location
Education	Degree, institution, specialization, graduation year
Experience	Company, role, duration, responsibilities

Category	Examples
Projects	Project purpose, contribution, technologies, outcome
Skills	Languages, frameworks, databases, tools, AI/ML skills
Career Preference	Desired roles, domains, work preferences
Evidence	Projects, experience, certifications or demonstrated knowledge

The system calculates a completeness score and identifies information that is missing, vague, contradictory or insufficient for meaningful role recommendations.

7. Module 4 - Intelligent Gap Filling

When important information is missing, the platform should not ask the candidate to fill a long static form. Instead, it should conduct a short conversational gap-filling process. Questions should be generated from the candidate's existing profile and should focus on information that meaningfully improves the profile.

- Ask only relevant questions.
- Avoid asking for information already present in the resume or previous answers.
- Prioritize high-value missing fields.
- Store each useful answer back into the structured profile.
- Make the newly supplied information available to the RAG layer where appropriate.
- Recalculate profile completeness after answers are processed.
- Stop when the required completeness threshold is reached or when no high-value questions remain.

8. Module 5 - Candidate Profile

The result of resume parsing and gap filling is a unified candidate profile. This is not a public portfolio website. It is the platform's structured representation of the candidate and acts as the primary source for role recommendations and assessments.

- Education profile.
- Experience timeline.
- Project profile.
- Skill inventory.
- Career interests and preferred roles.
- Evidence supporting claimed skills.
- Profile completeness.
- Assessment history and future benchmark metrics.

9. Module 6 - Job Role Recommendation

After the candidate profile becomes sufficiently complete, the system recommends roles rather than immediately recommending individual job openings. Recommendations should consider the complete candidate context instead of simple keyword matching.

- Education and specialization.
- Total and relevant experience.
- Projects and responsibilities.
- Demonstrated technical skills.
- Transferable skills.
- Candidate preferences.
- Estimated seniority and readiness.

The platform can present around five relevant roles, after which the candidate selects approximately two to three roles they are genuinely interested in pursuing. These selected roles become the targets for the adaptive assessment and later improvement process.

10. Module 7 - Adaptive Role Assessment

For each selected role, the platform conducts an adaptive assessment. Questions must not be a fixed questionnaire. Every next question should consider the previous question, the candidate's answer, demonstrated knowledge and the target role.

Adaptive difficulty logic

- If an answer is correct, detailed and confident, increase the depth or difficulty of the next question.
- If an answer is partially correct, ask a related question that verifies the missing concept.
- If an answer shows weak fundamentals, reduce difficulty and test foundational knowledge.
- Use follow-up questions to distinguish memorized answers from actual understanding.
- Cover enough competencies to make the final benchmark meaningful.
- Do not judge the candidate from one answer alone.

Example: if a candidate targeting a backend role clearly explains REST APIs, the next question can move toward authentication, database transactions, caching, asynchronous processing or system design. If the candidate struggles with REST fundamentals, the system should first test simpler API concepts before increasing difficulty.

11. Module 8 - Candidate Metrics and Benchmark

The assessment results are converted into a candidate benchmark. The benchmark should describe both the candidate's current level and the gap between the current level and the selected target roles.

- Technical fundamentals.
- Role-specific knowledge.
- Practical implementation ability.
- Project depth.
- Problem solving.
- Conceptual clarity.
- Communication / explanation quality where relevant.

- Profile evidence strength.
- Overall role readiness.

The platform may classify readiness into understandable levels such as Foundation, Entry-Level Ready, Strong Entry-Level, Intermediate Potential, or Needs Improvement. Exact labels and scoring thresholds should be calibrated using assessment data rather than treated as permanent fixed rules.

12. Core Product Idea - Raise the Candidate Benchmark

The main purpose of the platform is not simply to score candidates. The benchmark becomes the starting point for improvement. If a candidate wants a particular role but currently falls below the expected level, the system identifies the specific gaps preventing readiness.

Future stages can then recommend learning tasks, project work, practice assessments and other interventions. The candidate can be reassessed over time so that the platform measures whether the benchmark is actually improving.

13. Suggested High-Level Architecture

Layer	Responsibility
Candidate Application	Login, resume upload, profile, role selection, assessment UI
Backend API	Authentication, candidate data, workflow orchestration
Resume Processing	Document extraction, normalization and structured parsing
AI Layer	Gap questions, role reasoning, adaptive questions, evaluation
Relational Database	Authoritative candidate profile and assessment records
Vector Database	Resume/profile embeddings and retrieval
RAG Service	Retrieve candidate evidence for AI operations
Scoring Engine	Completeness, assessment and readiness metrics

14. Phase 1 Deliverables

- Candidate authentication and basic account flow.
- Resume upload endpoint and storage.
- Resume text extraction and structured parsing.
- Candidate profile schema.
- RAG ingestion and candidate retrieval.
- Profile completeness and missing-field detection.
- Conversational gap-filling workflow.
- Candidate profile read/update functionality.
- Role recommendation engine.
- Candidate role selection.
- Adaptive question generation.

- Answer evaluation and next-question difficulty adjustment.
- Candidate benchmark and readiness report.
- API documentation and basic testing.

15. Out of Scope for Phase 1

- Full recruiter portal.
- Automatic job applications.
- Large-scale external job crawling.
- Interview scheduling.
- Offer management.
- Payroll or employee onboarding.
- Advanced learning management functionality.

16. Expected Phase 1 Result

At the end of Phase 1, a candidate should be able to enter the platform with only a resume, complete missing information through an AI-guided conversation, receive a structured profile, discover suitable job-role directions, select preferred roles, complete an adaptive assessment and receive a clear benchmark of current readiness.

This creates the foundation for later phases where the platform can actively help candidates improve their benchmark, match them with real opportunities and measure progress toward employability.

17. Simplified Pipeline

Resume -> Parse -> Structure -> RAG Ingestion -> Detect Gaps -> Ask Questions -> Complete Profile -> Recommend Roles -> Candidate Selects Roles -> Adaptive Assessment -> Benchmark -> Identify Improvement Areas